Ritik Singh

Full Stack Developer

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SUMMARY

Full-Stack Developer with hands-on experience building scalable web applications, real-time platforms, and AI-powered systems. Proficient across the stack using **React.js, Node.js, Python, FastAPI, MongoDB, WebSocket, and AWS**. Skilled in engineering end-to-end solutions, from responsive frontends and robust backend architectures to database design, cloud deployment, and AI workflows. Passionate about solving complex, real-world problems through clean, maintainable, and user-focused technology.

INTERNSHIP

DelveX Innovations Pvt Ltd, *Full Stack Developer* 04/2026 – 10/2026 | Jaipur, Rajasthan

So basically, I was responsible as a Full-Stack Developer there, I built web applications from frontend UIs to backend microservices, databases, and cloud pipelines. My main project was Cloud Command Center—an AWS Infrastructure AI Assistant, which I engineered end-to-end.

What I Learned: • Mastered full-stack architectures and real-time state sync.
• Learned to bridge web apps with next-gen AI pipelines.
• Team Work.

EDUCATION

Bachelor of Computer Applications, *GLA University* 07/2023 – 09/2026 | Mathura, U.P.

Class XII, *Dhata Inter College* 04/2022 – 03/2023 | Fatehpur, U.P.

Class X, *Ch. P. S. High School* 04/2020 – 03/2021 | Fatehpur, U.P.

SKILLS

- MERN-Stack
- React.js
- Vue.js
- Next.js
- JavaScript
- HTML, CSS
- Bootstrap, Tailwind
- Node.js
- Express.js
- TypeScript
- Angular
- Python
- Java
- MongoDB, MySQL
- AWS
- Docker
- Terraform
- Linux
- Apache, NGINX
- Git, Github

Advanced Skills
- RAG
- VectorDB
- Agentic AI
- LLMs, NLP

PROJECTS

1. Jessy - AI Powered Voice Assistant 08/2026 – 10/2026

An advanced, full-stack AI-powered desktop voice assistant that bridges natural language processing with cross-platform automation. This intelligent system utilizes cutting-edge LLMs and real-time communication protocols to listen, reason, speak, and execute complex workflows across both local desktop computers and connected Android devices via hands-free voice commands.

Key Features & Technical Architecture:

- **Voice-First Pipeline:** Built a low-latency speech processing engine featuring offline **wake-word detection**, automated speech recognition via **Whisper**, and dynamic speech synthesis using a local/cloud **Text-to-Speech (TTS)** engine.
- **LLM Reasoning & Orchestration:** Integrated **Groq LLM** to interpret multi-turn conversational context, handle complex reasoning, and map open-ended natural language intent into structured system actions.
- **Real-Time Control Center:** Engineered a high-performance **FastAPI** backend that utilizes **WebSockets** for bidirectional, real-time data streaming to a responsive **React.js** dashboard—displaying live status, transcription streams, and current tasks.
- **Cross-Platform Device Automation:** Implemented deep system automation hooks enabling local computer control (file handling, application launching) alongside direct Android phone management using **ADB** and **UIAutomator2** to automate phone calls, contacts, and application navigation like YouTube.

2. A Chat Application , *MERN Stack based* 05/2026 – 07/2026

- Developed a **full-stack real-time chat application** leveraging the **MERN stack** and **Socket.io** for low-latency, bidirectional communication.
- Implemented secure user authentication processes including **Signup and Login** functionalities utilizing **JWT** and password hashing.
- Designed a dynamic user directory enabling seamless text-based connectivity among all registered platform users.

3. iQuiz - A Quiz Platform like Kahoot! , *MERN Stack based* 11/2025 – 01/2026

An interactive, full-stack multiplayer live-quiz platform that enables real-time engagement through dynamic game sessions and live leaderboard tracking. Built with a focus on low-latency synchronization, the platform allows hosts to generate unique quiz sessions while participants seamlessly join and compete instantly via a secure PIN system.

Key Features:

- **PIN-Based Room Management:** Secure, dynamic session handling that allows players to instantly join specific live-hosted lobbies using a unique session PIN.
- **Real-Time Data Sync:** Powered by Socket.io to handle instantaneous participant interactions, live question transitions, and low-latency scoreboard delivery.
- **Full CRUD Management:** Comprehensive dashboard for creators to **Create, Read, Update, and Delete** custom quizzes, question banks, and answer choices.
- **Dynamic Live Leaderboard:** An automated scoring engine that instantly calculates player ranks and broadcasts live leaderboard updates to all connected screens mid-game.
- **Stateful Host Controls:** A dedicated hosting portal providing full session control to start games, advance questions, and manage active participant lists.